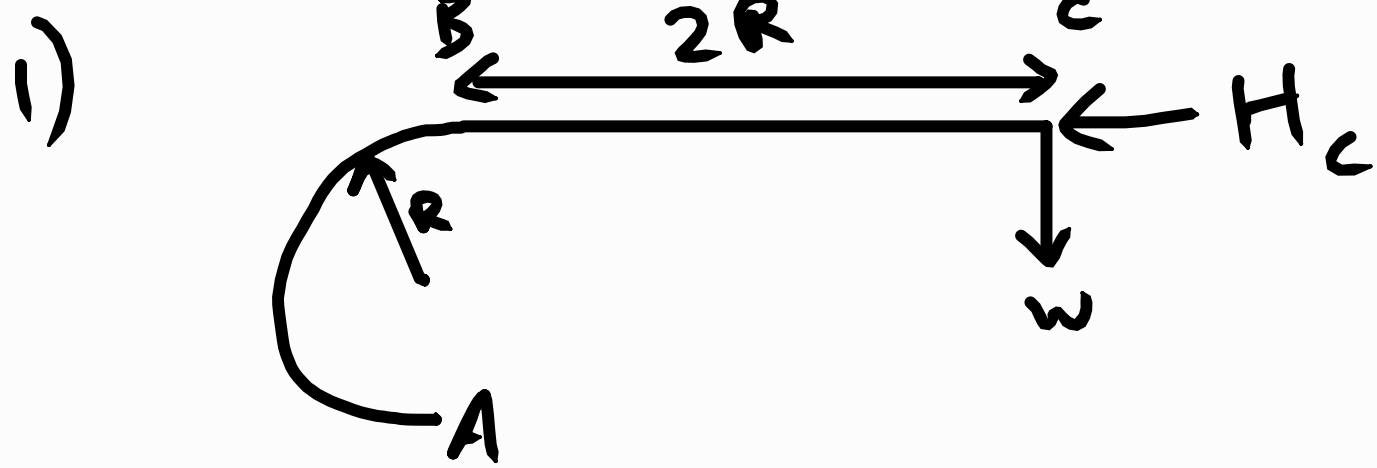
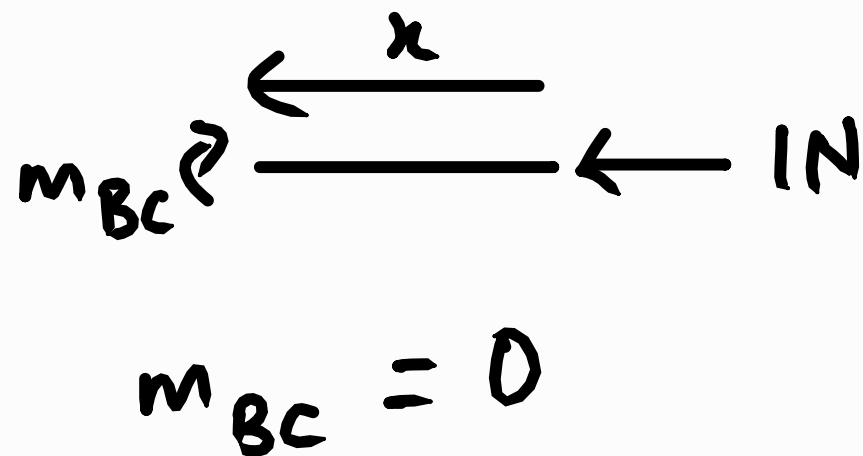
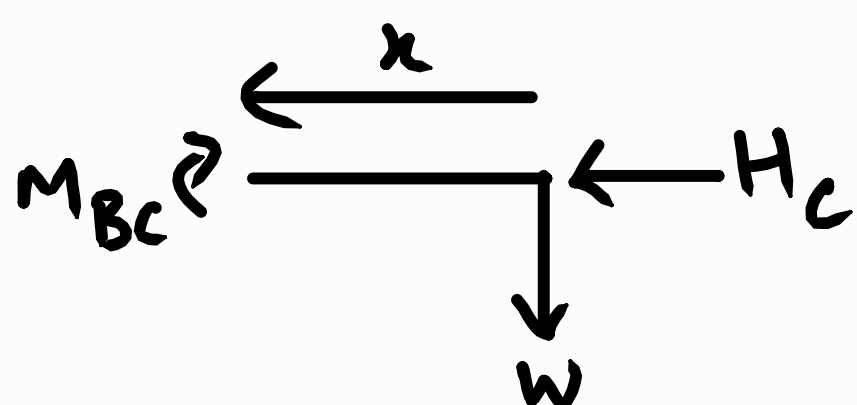




$$\Delta = \int_0^{2R} \frac{M_{BC} m_{BC}}{EI} dx + \int_0^{\pi} \frac{M_{AB} m_{AB} R}{EI} d\theta$$

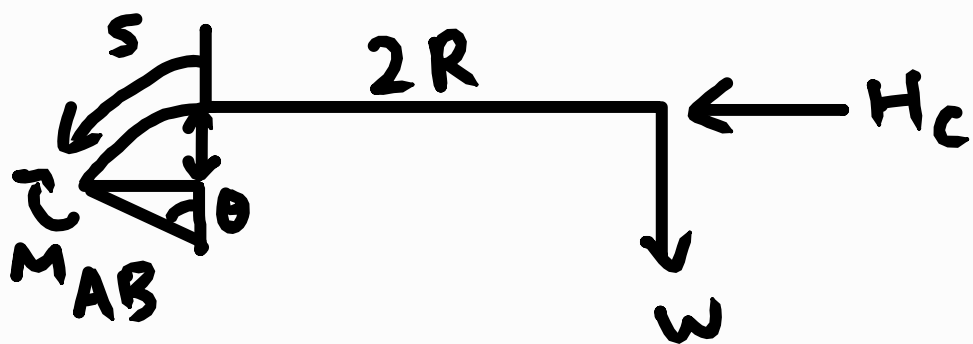
Segment BC:



$$M_{BC} + wx = 0$$

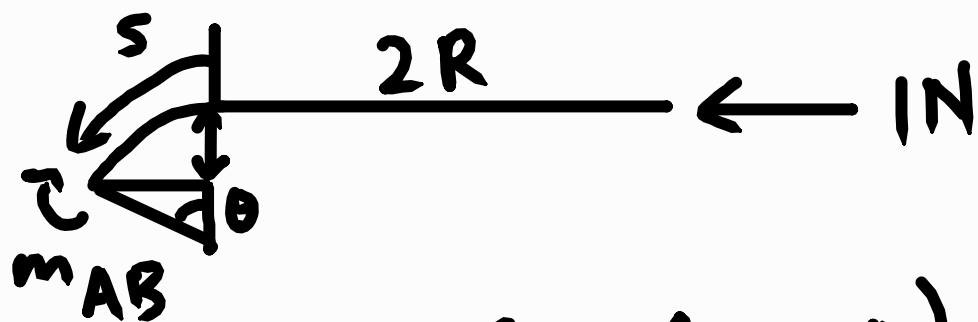
$$M_{BC} = -wx$$

Segment AB:



$$M_{AB} + w(2R + R \sin \theta) = H_c(R - R \cos \theta)$$

$$M_{AB} = H_c R(1 - \cos \theta) - wR(2 + \sin \theta)$$



$$m_{AB} = 1(R - R \cos \theta)$$

$$= R(1 - \cos \theta)$$

$$1) \therefore M_{AB} = H_c R(1 - \cos \theta) - WR(2 + \sin \theta)$$

$$m_{AB} = R(1 - \cos \theta)$$

$$M_{BC} = -Wx$$

$$m_{BC} = 0$$

$$\Delta = \int_0^{2R} \frac{M_{BC} m_{BC}}{EI} dx + \int_0^\pi \frac{M_{AB} m_{AB}}{EI} R d\theta$$

$$0 = \int_0^\pi \frac{[H_c R(1 - \cos \theta) - WR(2 + \sin \theta)][R(1 - \cos \theta)]}{EI} R d\theta$$

$$0 = \int_0^\pi \frac{R^3 [H_c(1 - \cos \theta) - W(2 + \sin \theta)](1 - \cos \theta)}{EI} d\theta$$

$$0 = \frac{R^3}{EI} \left[H_c \int_0^\pi (1 - \cos \theta)^2 d\theta - W \int_0^\pi (2 + \sin \theta)(1 - \cos \theta) d\theta \right]$$

$$0 = \frac{3}{2} \frac{R^3}{EI} H_c \pi - \frac{WR^3}{EI} \int_0^\pi 2 - 2 \cos \theta + \sin \theta - \sin \theta \cos \theta d\theta$$

$$0 = \frac{3}{2} \frac{R^3}{EI} H_c \pi - \frac{WR^3}{EI} [2\pi + 2]$$

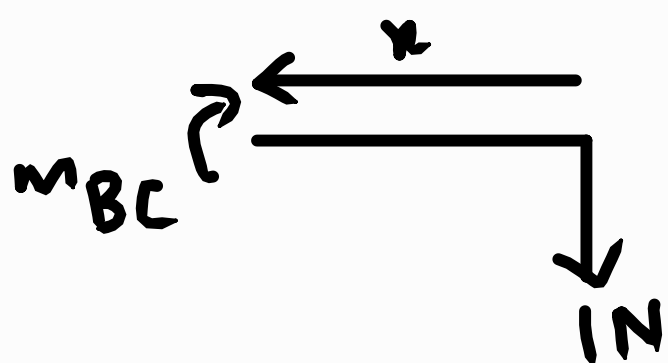
$$\frac{WR^3}{EI} [2\pi + 2] = \frac{3R^3 \pi}{2EI} H_c$$

$$2W[\pi + 1] = \frac{3\pi}{2} H_c$$

$$H_c = \frac{4W}{3\pi} (\pi + 1)$$

$$1) \Delta = \int_0^{2R} \frac{M_{BC} m_{BC}}{EI} dx + \int_0^{\pi} \frac{M_{AB} m_{AB}}{EI} R d\theta$$

Segment BC:
Vertical unit load

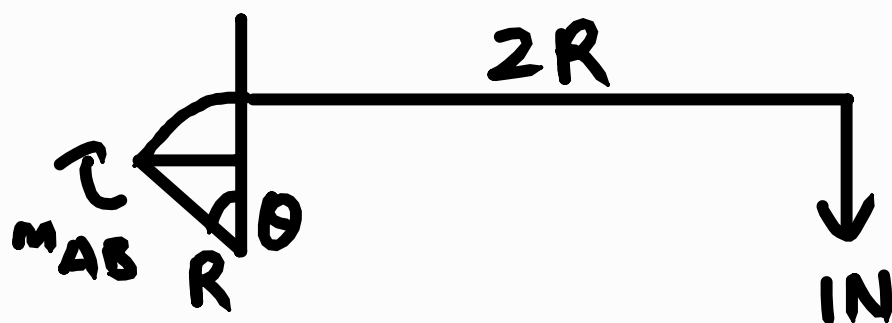


$$M_{BC} + x = 0$$

$$m_{BC} = -x$$

Segment AB:

Vertical unit load



$$m_{AB} + 1(2R + R \sin \theta) = 0$$

$$= -R(2 + \sin \theta)$$

$$\therefore M_{AB} = H_c R(1 - \cos \theta) - WR(2 + \sin \theta)$$

$$m_{AB} = -R(2 + \sin \theta)$$

$$M_{BC} = -wx$$

$$m_{BC} = -x$$

$$\Delta = \int_0^{2R} \frac{M_{BC} m_{BC}}{EI} dx + \int_0^{\pi} \frac{M_{AB} m_{AB}}{EI} R d\theta$$

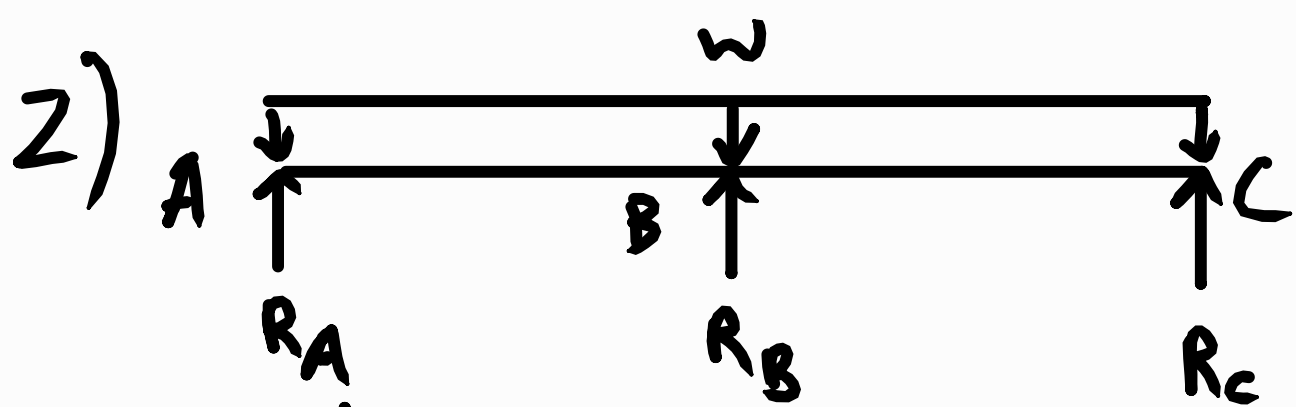
$$= \int_0^{2R} \frac{-wx(-x)}{EI} dx + \int_0^{\pi} \frac{[H_c R(1 - \cos \theta) - WR(2 + \sin \theta)][-R(2 + \sin \theta)] R d\theta}{EI}$$

$$= \frac{w}{EI} \left[\frac{x^3}{3} \right]_0^{2R} + \frac{R^3}{EI} \int_0^{\pi} w(2 + \sin \theta)^2 - H_c(1 - \cos \theta)(2 + \sin \theta) d\theta$$

$$= \frac{8WR^3}{3EI} + \frac{R^3}{EI} \int_0^{\pi} w(2 + \sin \theta)^2 - \frac{4W}{3\pi} (\pi + 1)(1 - \cos \theta)(2 + \sin \theta) d\theta$$

$$= \frac{8WR^3}{3EI} + \frac{WR^3}{EI} \int_0^{\pi} (2 + \sin \theta)^2 - \frac{4}{3\pi} (\pi + 1)(1 - \cos \theta)(2 + \sin \theta) d\theta$$

$$= 10.2440935 \left(\frac{WR^3}{EI} \right) \approx 10.24 \left(\frac{WR^3}{EI} \right)$$



$$\Delta = \int_0^{\frac{L}{2}} \frac{M_{AB} m_{AB}}{EI} dx + \int_0^{\frac{L}{2}} \frac{M_{BC} m_{BC}}{EI} dx$$

$$\sum M_A: R_B \left(\frac{L}{2} \right) + R_C (L) = wL \left(\frac{L}{2} \right)$$

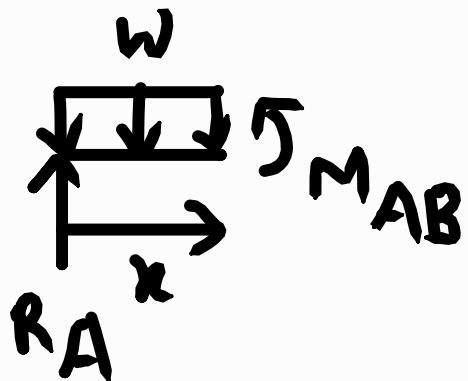
$$R_C = \frac{1}{2} (wL - R_B)$$

$$\sum M_C: R_A L + R_B \left(\frac{L}{2} \right) = wL \left(\frac{L}{2} \right)$$

$$R_A = \frac{1}{2} (wL - R_B)$$

$$\therefore R_C = R_A = \frac{1}{2} (wL - R_B)$$

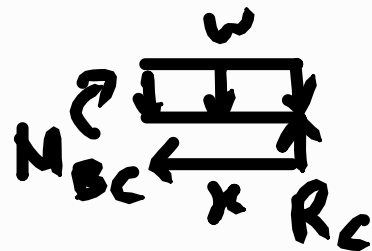
Segment AB:



$$M_{AB} + wx \left(\frac{x}{2} \right) = R_A x$$

$$M_{AB} = \frac{1}{2} x (wL - R_B) - \frac{wx^2}{2}$$

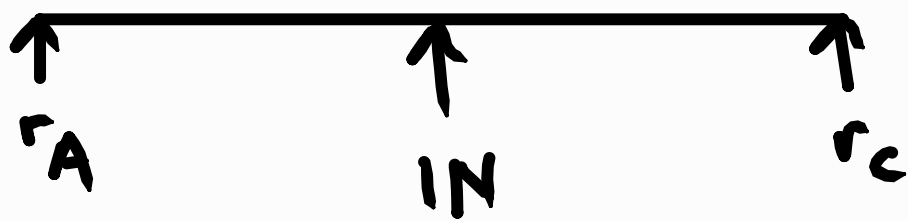
Segment BC:



$$M_{BC} + wx \left(\frac{x}{2} \right) = R_C x$$

$$M_{BC} = \frac{1}{2} x (wL - R_B) - \frac{wx^2}{2}$$

2) Virtual force analysis:



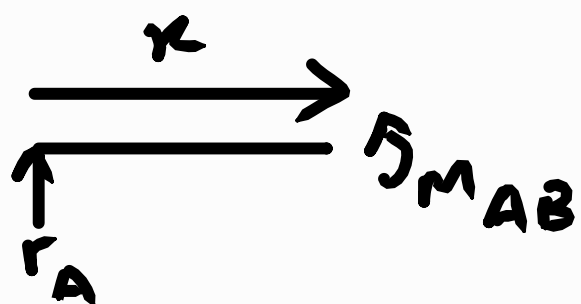
$$\sum m_A: 1\left(\frac{L}{2}\right) + r_c L = 0$$

$$r_c = -\frac{1}{2}$$

$$\sum m_c: 1\left(\frac{L}{2}\right) + r_A L = 0$$

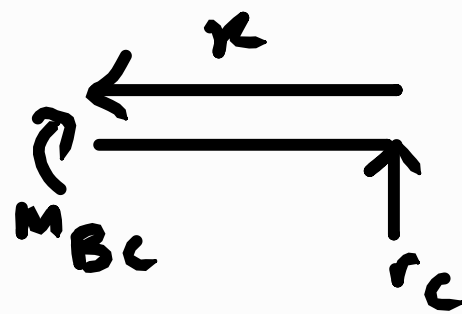
$$r_A = -\frac{1}{2}$$

Segment AB:



$$m_{AB} = r_A x = -\frac{1}{2}x$$

Segment BC:



$$m_{BC} = r_c x = -\frac{1}{2}x$$

$$\therefore M_{AB} = M_{BC} = \frac{1}{2}x(wL - R_B) - \frac{wx^2}{2}$$

$$m_{AB} = m_{BC} = -\frac{1}{2}x$$

$$\therefore \Delta = \int_0^{\frac{L}{2}} \frac{M_{AB} m_{AB}}{EI} dx + \int_0^{\frac{L}{2}} \frac{M_{BC} m_{BC}}{EI} dx$$

$$= 2 \int_0^{\frac{L}{2}} \frac{M_{AB} m_{AB}}{EI} dx$$

$$= \frac{2}{EI} \int_0^{\frac{L}{2}} \left[\frac{1}{2}x(wL - R_B) - \frac{wx^2}{2} \right] \left[-\frac{1}{2}x \right] dx$$

$$\begin{aligned}
 2) \Delta &= \frac{2}{EI} \int_0^{\frac{L}{2}} \left[\frac{1}{2}x(wL - R_B) - \frac{wx^2}{2} \right] \left[-\frac{1}{2}x \right] dx \\
 &= \frac{1}{2EI} \int_0^{\frac{L}{2}} -x \left[x(wL - R_B) - wx^2 \right] dx \\
 &= \frac{1}{2EI} \int_0^{\frac{L}{2}} -x^2(wL - R_B) + wx^3 dx \\
 &= \frac{1}{2EI} \int_0^{\frac{L}{2}} R_B x^2 - wL x^2 + wx^3 dx \\
 &= \frac{1}{2EI} \left[\frac{R_B x^3}{3} - \frac{wL x^3}{3} + \frac{wx^4}{4} \right]_0^{\frac{L}{2}} \\
 &= \frac{1}{2EI} \left[\frac{R_B L^3}{24} - \frac{wL^4}{24} + \frac{wL^4}{64} \right] \\
 &= \frac{1}{2EI} \left[\frac{R_B L^3}{24} - \frac{5}{192} wL^4 \right]
 \end{aligned}$$

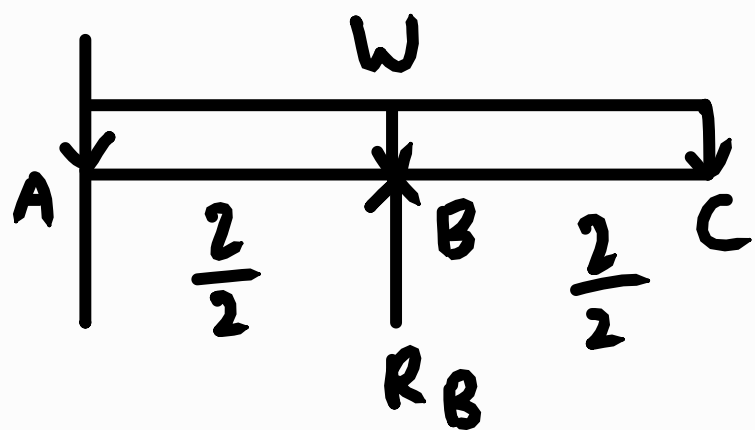
Since $\Delta_B = 0$,

$$\therefore 0 = \frac{1}{2EI} \left[\frac{R_B L^3}{24} - \frac{5}{192} wL^4 \right]$$

$$\frac{R_B L^3}{24} = \frac{5}{192} wL^4$$

$$R_B = \frac{5}{8} wL$$

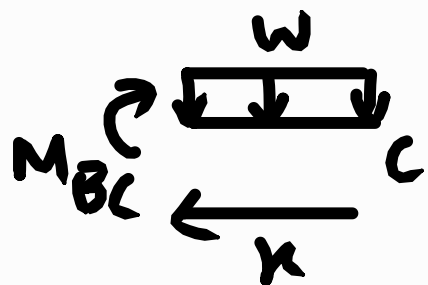
3)



Spring
 $\uparrow R_B$
 $\downarrow R_B$
 $\approx$
 $\uparrow R_B$

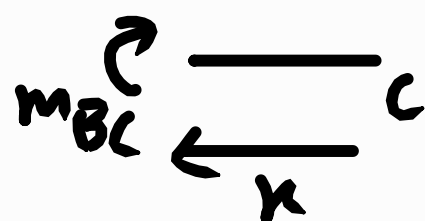
$$\Delta = \int_0^{\frac{l}{2}} \frac{M_{AB} M_{AB}}{EI} dx + \int_0^{\frac{l}{2}} \frac{M_{BC} M_{BC}}{EI} dx + \frac{Ff}{k}$$

Segment BC:



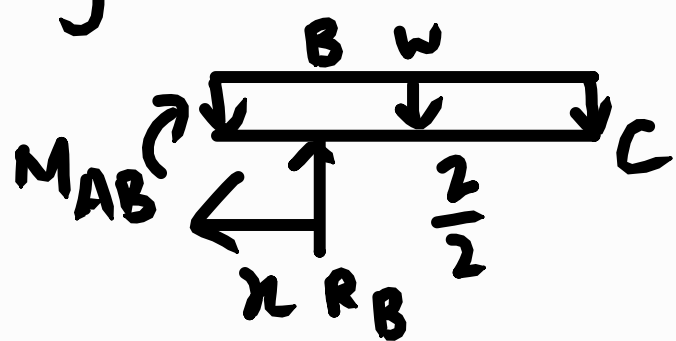
$$M_{BC} + wx \left(\frac{x}{2} \right) = 0$$

$$M_{BC} = -\frac{wx^2}{2}$$



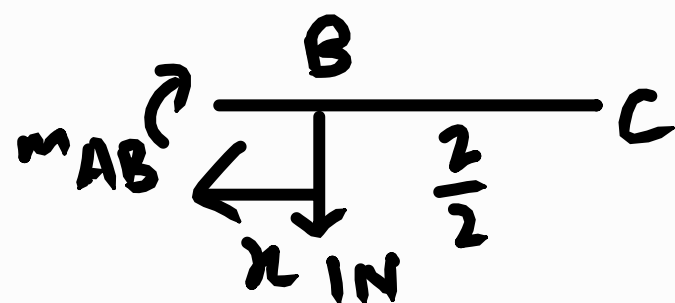
$$m_{BC} = 0$$

Segment AB:



$$M_{AB} + w \left(x + \frac{l}{2} \right) \left(\frac{1}{2} \right) \left(x + \frac{l}{2} \right) = R_B x$$

$$M_{AB} = R_B x - \frac{1}{2} w \left(x + \frac{l}{2} \right)^2$$



$$m_{AB} = -x$$

$$3) \therefore M_{BC} = -\frac{wx^2}{2}, m_{BC} = 0, M_{AB} = R_B x - \frac{1}{2}w\left(x + \frac{l}{2}\right)^2, m_{AB} = x,$$

$$F = -F_B, f = 1, k = \frac{8EI}{l^3}$$

$$\Delta = \int_0^{\frac{l}{2}} \frac{M_{BC} m_{BC}^0}{EI} dx + \int_0^{\frac{l}{2}} \frac{M_{AB} m_{AB}}{EI} dx + \frac{Ff}{k}$$

$$= \frac{1}{EI} \int_0^{\frac{l}{2}} \left(R_B x - \frac{1}{2}w\left(x + \frac{l}{2}\right)^2 \right) (-x) dx + \frac{-R_B(1)}{k}$$

$$= \frac{1}{EI} \int_0^{\frac{l}{2}} R_B x^2 - \frac{1}{2}wx\left(x + \frac{l}{2}\right)^2 dx - \frac{R_B}{k}$$

$$= -\frac{R_B}{k} + \frac{1}{EI} \int_0^{\frac{l}{2}} -R_B x^2 + \frac{1}{2}wx\left(x^2 + 2x + \frac{l^2}{4}\right) dx$$

$$= -\frac{R_B}{k} + \frac{1}{EI} \int_0^{\frac{l}{2}} -R_B x^2 + \frac{w}{2}x^3 + \frac{wl}{2}x^2 + \frac{wl^2}{8}x dx$$

$$= -\frac{R_B}{k} + \frac{1}{EI} \left[-\frac{1}{3}R_B x^3 + \frac{w}{8}x^4 + \frac{wl}{6}x^3 + \frac{wl^2}{16}x^2 \right]_0^{\frac{l}{2}}$$

$$= -\frac{R_B}{k} + \frac{1}{EI} \left[-\frac{R_B l^3}{24} + \frac{w}{128}l^4 + \frac{wl}{48}l^4 + \frac{wl^2}{64}l^4 \right]$$

$$= -\frac{R_B}{k} - \frac{R_B l^3}{24EI} + \frac{17}{384} \frac{wl^4}{EI}$$

$$= -\frac{R_B}{k} - \frac{R_B l^3}{24EI} + \frac{17wl^4}{384EI}$$

3) Since $\Delta_B = 0$,

$$0 = -\frac{R_B}{k} - \frac{R_B l^3}{24EI} + \frac{17wl^4}{384EI}$$

$$0 = -\frac{R_B}{\frac{8EI}{l^3}} - \frac{R_B l^3}{24EI} + \frac{17wl^4}{384EI}$$

$$\frac{R_B l^3}{6EI} = \frac{17wl^4}{384EI}$$

$$R_B = \frac{17}{64} wl$$

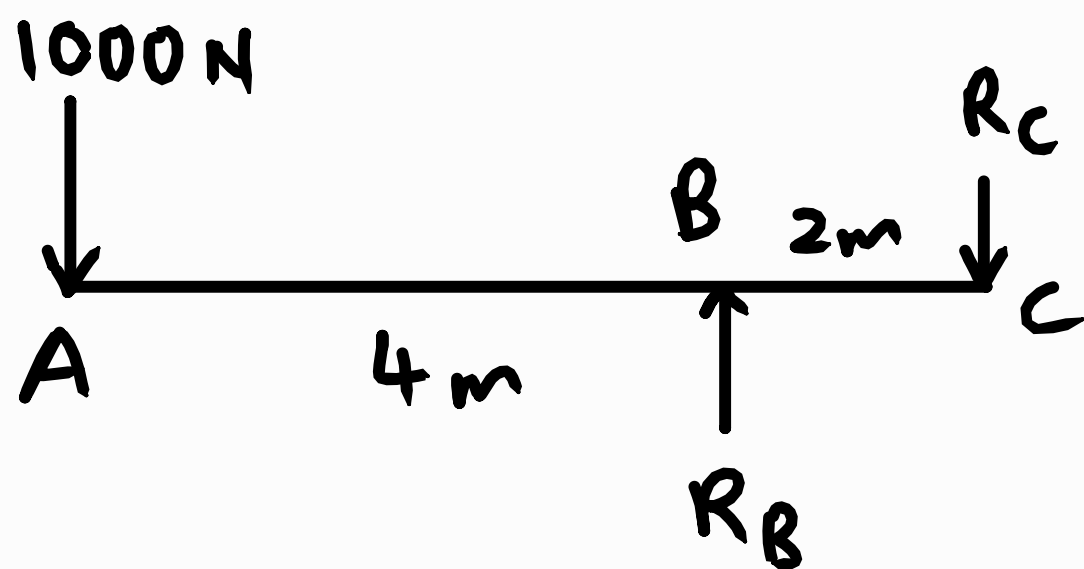
When the support is rigid, $k \rightarrow \infty$

$$\Delta = -\frac{R_B}{k} - \frac{R_B l^3}{24EI} + \frac{17wl^4}{384EI}$$

$$\frac{R_B l^3}{24EI} = \frac{17wl^4}{384EI}$$

$$R_B = \frac{17}{16} wl$$

4)



Let Δ_s be the static deflection at tip of the spring board with the weight applied gradually.

To find the maximum deflection,

$$\Delta_{\max} = \Delta_s \left(1 + \sqrt{1 + \frac{2h}{\Delta_s}} \right)$$

$$\sum M_c = 0$$

$$1000(6) = 2R_B$$

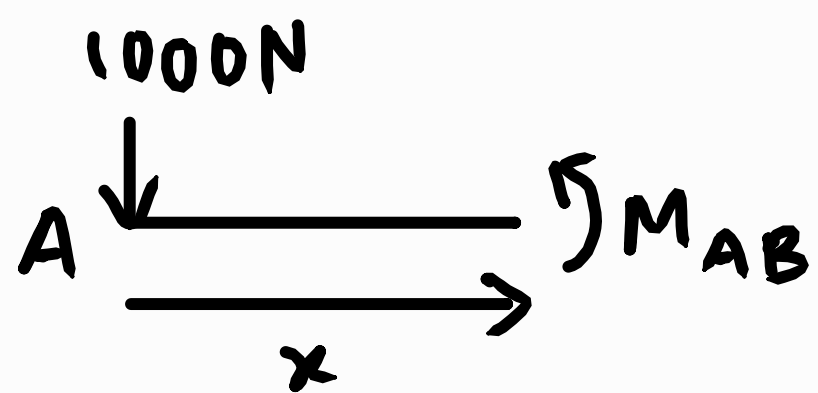
$$R_B = 3000$$

$$\sum F_y = 0$$

$$R_B = 1000 + R_C$$

$$R_C = 3000 - 1000 = 2000$$

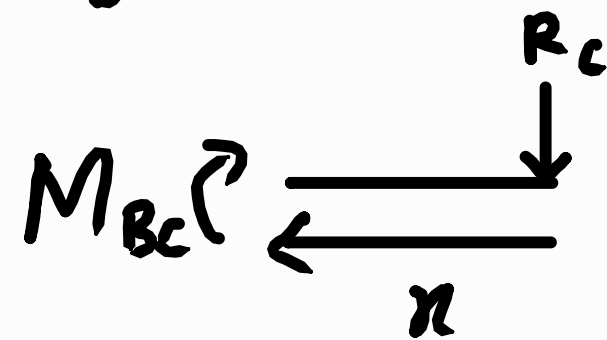
4) Segment AB:



$$M_{AB} + 1000x = 0$$

$$M_{AB} = -1000x$$

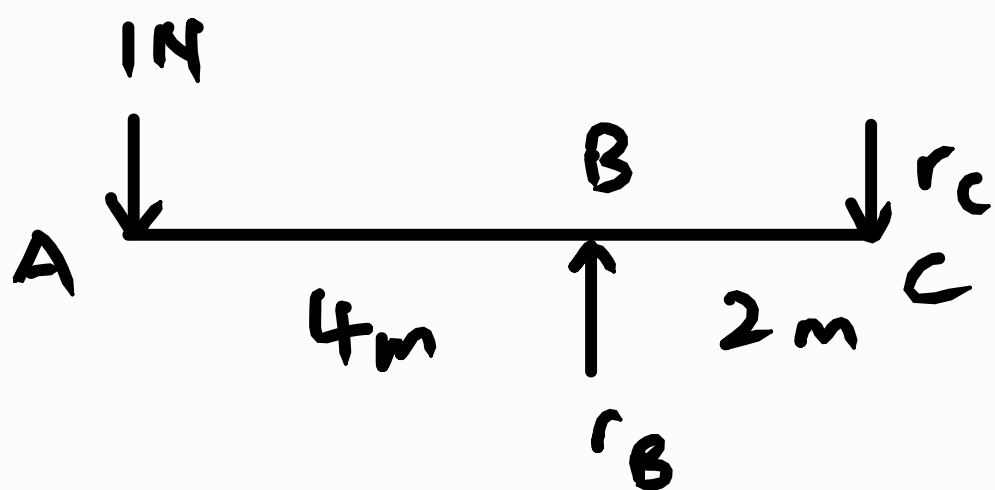
Segment BC:



$$M_{BC} + R_c x = 0$$

$$M_{BC} = -R_c x$$
$$= -2000x$$

Virtual load analysis



$$\sum M_c = 0$$

$$1(6) = 2r_B$$

$$r_B = 3 \text{ N}$$

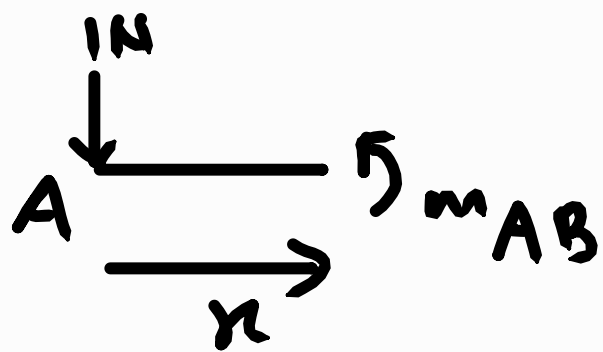
$$\sum F_y = 0$$

$$1 + r_c = r_B$$

$$r_c = r_B - 1$$

$$= 3 - 1 = 2 \text{ N}$$

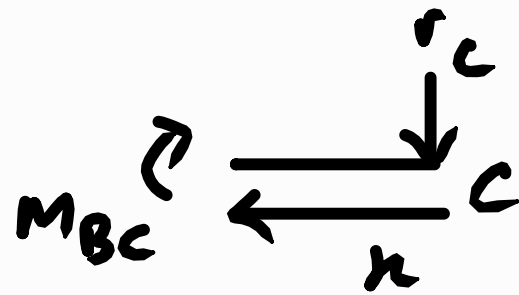
4) Segment AB:



$$m_{AB} + 1x = 0$$

$$m_{AB} = -x$$

Segment BC:



$$m_{BC} + r_c x = 0$$

$$m_{BC} = -r_c x \\ = -2x$$

$$\therefore M_{AB} = -1000x, m_{AB} = -x, M_{BC} = -2000x, m_{BC} = -2x$$

$$\Delta_s = \int_0^4 \frac{-1000x(-x)}{EI} dx + \int_0^2 \frac{-2000x(-2x)}{EI} dx$$

$$= \frac{1}{EI} \int_0^4 1000x^2 dx + \frac{1}{EI} \int_0^2 4000x^2 dx$$

$$= \frac{1}{EI} (32000)$$

$$= \frac{1}{160 \times 10^3} (32000)$$

$$\approx 0.2 \text{ m}$$

$$\Delta_{max} = \Delta_s \left(1 + \sqrt{1 + \frac{2h}{\Delta_s}} \right)$$

$$= 0.2 \left(1 + \sqrt{1 + \frac{2(1)}{0.2}} \right)$$

$$= \frac{1 + \sqrt{11}}{5} \approx 0.86 \text{ m}$$